

Course: Special Ship Automation Systems 2026
Homework Assignment (HW)
System Modeling Using Experimental
Measurement Data (Part 1)

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1 The Problem

The experimental hybrid diesel-electric propulsion installation of the HIPPO-2 facility at the Laboratory of Marine Engineering (LME) consists of a Diesel engine with a maximum power of 261 kW and an electric motor with a power of 90 kW, which are connected to the same shaft, with an electric brake used to apply load to the system. The HIPPO-2 installation and the parameters measured during its operation are shown in Fig. 1. During experiments, numerous measurements are taken, which, although very useful for monitoring, fault diagnosis, etc., are quite difficult to obtain during the operation of an actual propulsion system. To achieve these goals, virtual sensors are often proposed as a solution, working in parallel with the other sensors of the installation. Additionally, we often aim to create a digital twin of the experimental setup or actual propulsion system to simulate the machine under various operating conditions and perform diagnostic checks. These requirements—virtual sensors and digital twins—can be addressed using neural network models.

2 Tasks to Solve

In this assignment, you are asked to develop such neural networks by following these steps:

1. Analyze the data using various graphical representations.
2. Filter and transform the data.
3. Split the data into training, validation, and test sets.
4. Develop **2 simple neural networks** to predict two of the following parameters (one network per parameter):
 - (a) NO_x,
 - (b) Fuel Consumption,

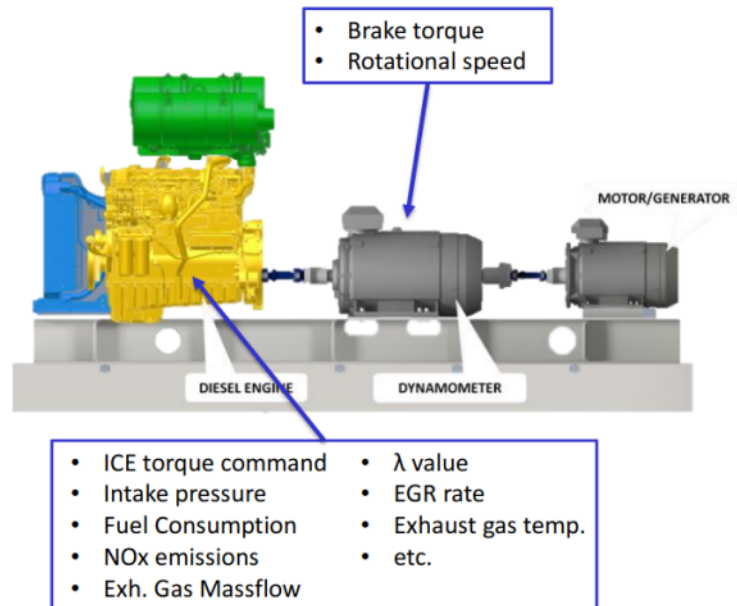


Figure 1: HIPPO-2

(c) Intake Pressure,

(d) λ value

using a combination of the remaining parameters (excluding the four being predicted).

5. Evaluate the results on the test set and provide plots of the training progress (Training loss vs. iteration, Validation loss vs. iteration).

3 Notes:

1. The data for the assignment can be found on the course website on Helios.
2. For step 4, use **1 hidden layer** with at most **20 neurons**. Create one network per parameter.
3. All plots must include **axis labels, curve labels, grids, and must not be screenshots** (use tips from the following repo *BeautifulFigures* as well as the videos in the author's channel *Principles of Beautiful Figures for Research Papers*).
4. Code must be in **Python**.
5. Code must include comments explaining the process.
6. Using Git and WandB will be taken positively into consideration.
7. Presentations must be between **10-15 slides** (for part 1)
8. For the presentation, use the template provided in Helios and follow the rules from the **How to present.pdf** in the extra material folder, as well as the general ideas from the *How to Speak* video.
9. The first part of the assignment will be presented around 15-22/5/2025.

References

- [1] L. Biewald, "Experiment Tracking with Weights and Biases," 2020. [Software]. Available: wandb.ai.